Kuaotunu Wildfire:

Case Study 5

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EnvSci 704: Modelling of Environmental Systems

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Executive Summary

This report simulates how vegetation density within a forest impacts the spread and self-extinguishing of a wildfire. Using NetLogo's agent-based modeling system, a model was created. The west border of the model simulates a road where the ignition source is located. The model is on a slope rising to the east, and the vegetation densities within are separated into four vertical sections. Densities were tested from 36% to 42% in one-percentage-point steps, and a sensitivity analysis was replicated for each step, totalling 70 runs. This model was created in hopes of creating a mitigation solution that reduces roadside ignition-point wildfires while simultaneously maintaining habitat for native flora and fauna.

Main Findings

Fire spread response to vegetation density has a threshold. At 36% vegetation density, 6.3% of the bush burned, staying wholly within the first sector, Section 0. At 40% vegetation density, an average of 88.1% burned, and at 42% vegetation density, 94.9% burned. However, at 38% vegetation density, 21.1% burned, with two instances of the fire leaving Section 0 burning around 75% of the vegetation. At 39% density, 39.7% burned, with four instances of above 80% burned. 38% and 39% showcase the slight unpredictability of the transition threshold.

Risk Reduction Mitigation

The findings from the model imply that keeping vegetation near potential ignition sources, such as roads, below 38%. This allows the fire time to self-extinguish before reaching the denser sections of the forest further from the ignition source.

Introduction/Brief

Roadside berm maintenance can be a costly service, but as climate change continues to increase dry spells, it might just become a necessity in reducing wildfires. This model examines how the density of vegetation affects the percolation threshold, or critical value (p), in the spread of wildfire via a lattice agent-based fire model (Wilensky, 1997). By controlling the density of vegetation, especially near roads, common ignition points, further vegetation retains the parameters to support habitats for native fauna. As much of New Zealand's native flora and fauna are endemic, maintaining wild habitats is important not just from an ecological view but a cultural standpoint (Environment New Zealand 2007).

Below a critical value threshold, wildfires *fizzle,* meaning they remain small and die out before reaching a burn point of 50%; above this threshold, wildfires cluster and *spread* (Figure 2). The critical threshold of forests varies depending on variables specific to each forest. This model, however, takes a general and basic standpoint on wildfires, modeled after the January 2025 Coromandel shrub fire in Kuaotunu. Despite its simplicity, the model generates complex fire patterns showing the importance of berm maintenance and vegetation thinning.

Methods

The model is divided into four equal vertical sections (Figure 1). Base density *d* is the foundation for Section 1, Section 2, and Section 3, with density percent increasing d + 3, d + 5, and d + 7, respectively. The ignition point is the western border of the model, representing Kuaotunu's Black Jack Road. The fire has the capability to spread in eight directions; this includes diagonals. Each tick of the model represents the fire spreading through the patches, with chances of igniting neighboring patches being 50 + (slope * dx * 100), where dx +1 is eastward, 0 is north or south,

and -1 is west. Slope is set at 0.5; this pushes fire uphill. The run ends when burning ceases.

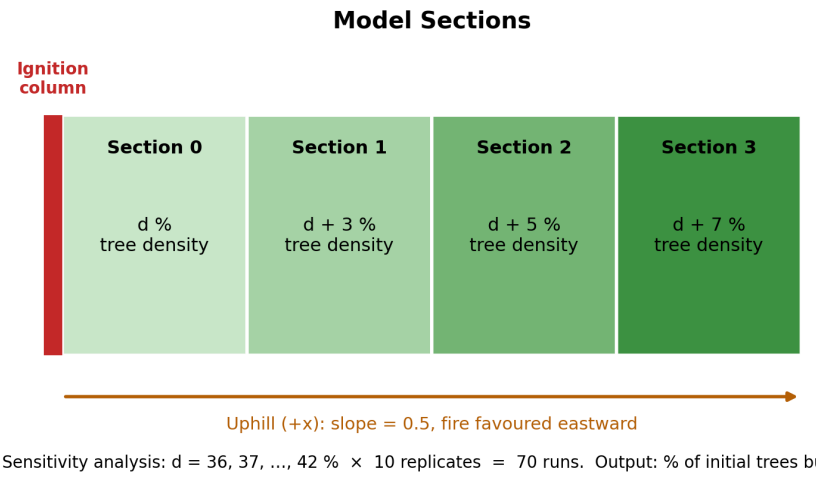


*Figure 1. Model Sections. Section 0 is the base density with Sections 1 (*d + *3), 2 (*d + *5), and 3 (*d + *7). The slope shows an east peak, favoring left-to-right fire spread.*

Everything, aside from *d,* remains constant; *d* varies by 1%, starting from 36% and ending at 42%. Model output is the percent of vegetation burned and a visual of burned vs. unburned patches. *Spread* signifies runs where over 50% of vegetation burned, and *fizzled* represents runs under 50% burned.

% Veg.	Run 1	Run 2	Run 3	Run 4	Run 5	Run 6	Run 7	Run 8	Run 9	Run 10	Mean
36	8.6	7.1	4.8	9.1	4.5	5.1	8.4	5.3	3.8	6.4	6.31
37	6.8	8.9	12.2	7.5	4.8	7.7	18.9	72.7	7.7	3.5	15.07
38	72.1	8.6	6.8	9	6.2	8.3	70.4	3.4	14.4	10.3	21.2
39	85.4	9.4	8.1	8	87.7	83.1	10.6	86.9	8.2	9.5	39.66
40	90.4	89.1	89.1	90.8	87.3	83.3	89.2	90.9	88	83.1	88.11
41	94	12.2	90.2	92.6	91.2	90.0	93.1	93.7	90.3	92.4	84.06
42	94.9	95.8	95	94.5	95.5	95.5	93.6	94.5	96	93.1	94.94

Table 1. Sensitivity Analysis. Shows the percent vegetation results and the mean over 10 runs.

A sensitivity analysis was conducted at each density percentage, 36-42%. Ten runs were run for each density setting, with the output percent burned being recorded and averaged (Table 1). The sensitivity analysis highlights the unpredictability of wildfire movement even in near-identical forests.
The code within the model added eight directional movements, slope, and four sections of varying density.

Results

Wildfire burning and spread are controlled by a threshold of a base density of 38% to 39%. Mean percent burned rose with vegetation density, with 36% vegetation having a burn of 6.31%, fizzling out within Section 0. 37% vegetation rose to 15.07% burned, with one case of the wildfire entering Section 1. Vegetation density of 38% is where variation begins, and the percolation threshold begins to surface, as the mean percent burned rose to 21.1% with the wildfire spreading to Sections 1, 2, and 3 in two distinct cases. At 39% vegetation, there are four cases of wildfire burning over 80% of the forest, raising the mean percent burned to 39%. 40% and 41% vegetation are within the 80% mean burned. 41% vegetation has a wildfire burn mean of 94.9%.

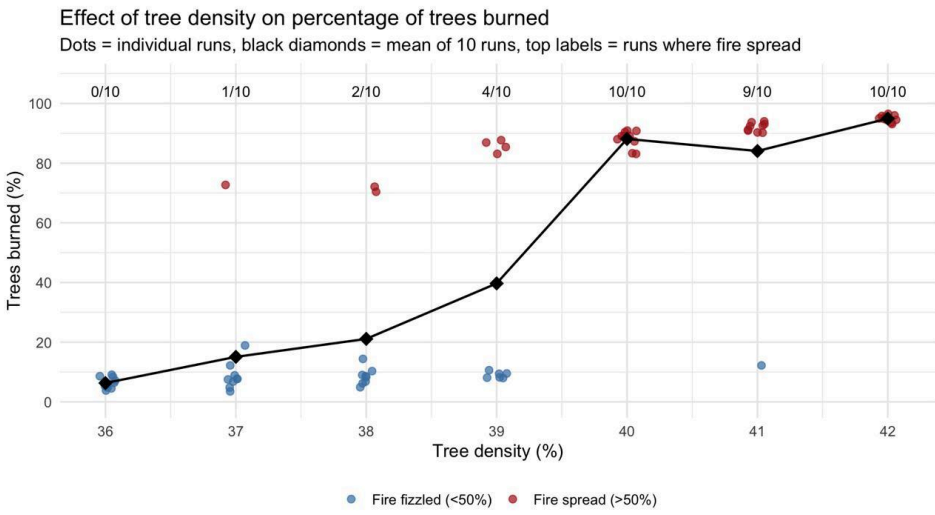


Figure 2. Percent of trees burned per run. Blue dots show the fizzled run, indicating less than 50% vegetation burned. Red dots indicate over 50% burned, and the black diamonds are the means. The top label shows runs with spread.

Figure 2 shows two outcome groups: fizzled and spread. Fizzled fires burned an average of 4-19%; fires that spread burned 70-96%. Interestingly, of the 70 runs, there were none between 20% and 69% burned. The progression from fizzled to spread is also documented in Figure 2, moving at 37% of 1 spread, 38% at 2, and 39% at 4. 41% outlier of fizzled at 12.2% burned showcases the unpredictability of wildfire movement even in near-identical forests.

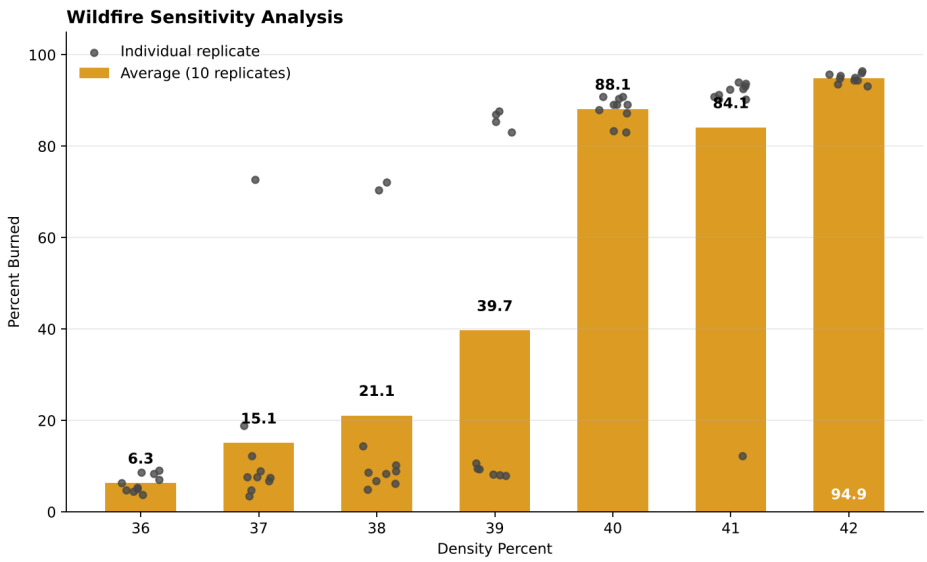


Figure 3. Wildfire Sensitivity Analysis. Bar graph of mean burned, showcasing 10 runs as dots.

Figure 3 shows averages and the percolation threshold. The mean between 39% and 40% vegetation density, from 39.7% to 88.1%, respectively, burned is a visual of the critical value of wildfire.

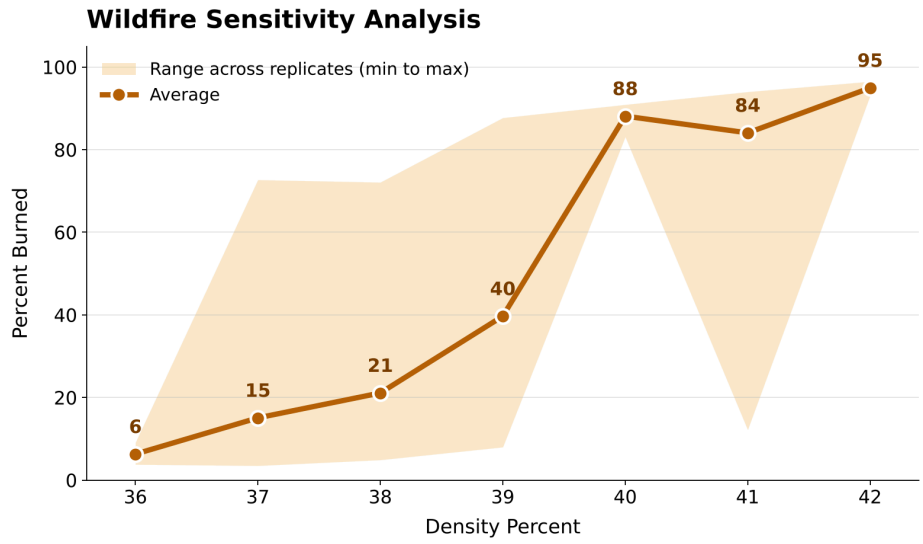


Figure 4. Wildfire Sensitivity Analysis. This figure showcases the same data as Figure 3, with the shaded data being the min and max range of the runs. The line is the mean.

This figure visualizes the unpredictability of wildfire spread, showcasing the possibility of fizzle even at percent densities above the critical threshold. As seen by the fizzle at 41%.
Wildfire spread behavior is unpredictable, but has controls that can be modified to reduce the risk to forests. The critical value, or percolation threshold, of the vegetation sits between 38% and 39%, as seen in the above figures. The jump in mean percent burned between these densities remained below 50%, which allows surviving vegetation surrounding the wildfire site ample space to support itself and fauna that might have fled the fire. Having vegetation that survived fires also permits a quicker recovery within the burned regions.

Discussion

The results support the mitigation strategy of thinning roadside vegetation to protect and reduce wildfire spread. This berm maintenance protects wilder vegetation further from the assumed ignition source, and this allows native flora to support fauna without the threat of roadside human-caused fire spread.

Sensitivity/Limitations

This is a simple agent-based model, created with the purpose of showcasing wildfire patterns over a varying range of vegetation densities. This model can be used as a baseline to build a mitigation strategy that has the potential to reduce bush burning as well as minimize wildfires as a whole.
Use of this model requires an understanding that wildfires are inconsistent and are not simple events. Every fire contains its own unique fuels, vegetation, and environmental parameters.

Meteorology

Meteorology is a major consideration in wildfire spread, and several of its factors are omitted from this model. Wind is notably absent from this model, as well as moisture. Wind not only fuels fire, but can dictate its movement. With its absence, it is important to understand that this model has a severe hindrance in practical use. Without moisture, how fast vegetation burns is unknown and thus places an assumption on the patch vegetation.

Ignition

The model assumes ignition point, the west border of the lattice and does not have a specific source. As this model is created using similar parameters to Kuaotunu's Black Jack Road wildfire, the ignition point is assumed to be a road source. This is not the case for all wildfires, as natural causes, such as lightning, can also be the point source.

Recommendations

The furthering of this model requires more variables. The addition of complex meteorological events, as well as creating desired point source ignitions, can make this model useful outside of berm maintenance and vegetation thinning. Their addition could be coded to Auckland's unique climate but also simplified to a basic universal fire model that adds either vegetation moisture or precipitation.
This model should also consider adding additional replicas of each percent vegetation, upping the 10 runs to 50+, this would allow for a deeper understanding and conclusion of wildfire thresholds, movements, and limitations.

Conclusion

Fire spread is controlled by a threshold. Maintaining roadside vegetation underneath this threshold, 38% to 39% reduces the chance of fire spread into deeper, denser vegetation. With vegetation density over 40%, fire spread is near absolute and has the potential to burn near 90% of an average site. By following this model and adding additional site-specific variables, a mitigation strategy of vegetation thinning has the capability of reducing wildfire risk. This not only protects against wildfire spread, but leaves deeper forest native habitats virtually untouched.

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